

Our Digital World — Paper D

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper. Every sum is built to work by hand, using small numbers or powers of ten.
- A label alone is rarely a full answer: where a question says **explain** or **describe**, give the action or the reason. A defence only scores with its reason — “install a firewall” earns nothing; say what it does that stops the attack.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe situations you may not have seen before. You are not expected to recognise them — apply the ideas you already have.

SECTION 1 – SORTING AND ORDERING

1. Four things from Aarav’s week. For each, write **A** or **P**, and give a short reason [4]
that says who put it there.

What happened	A or P	Reason
He uploaded a cricket video to his channel		
A coaching app logged where he was every hour		
A friend shared a screenshot of an old comment he had deleted		
He wrote a game review on a shop’s website, under his own name		

2. For each guard, state what it does **and** one thing it cannot do. A mark needs both. (a) Firewall [1]

- (b) Antivirus [1]

3. These are the five things to do when you are being cyberbullied, listed in the wrong order.

The step	Its place, 1 to 5
Tell a trusted adult	
Block the person	
Do not retaliate	
Report it to the platform and to school	
Save the evidence	

(a) Write 1 to 5 in the boxes to put the steps in the correct order. [2]

(b) Explain why “save the evidence” must come before “block the person”. [1]

SECTION 2 – EVIDENCE AND CALCULATION

4. A PIN is made only of digits. An attacker’s program makes **1,000 guesses every second**, so a 6-digit PIN takes 1,000 seconds to try in full. Deepak says: “Twice as many digits, so a 12-digit PIN takes twice as long: 2,000 seconds.”

(a) Write the number of 12-digit PINs as a power of ten. [1]

(b) Work out the time to try them all, in seconds, as a power of ten. [1]

(c) Taking one year as 3×10^7 seconds, estimate that time in years. [1]

(d) Explain Deepak’s mistake, in terms of what each extra digit does. [1]

5. Three posts arrive in a school group chat. The middle column of the table shows what a quick search finds for each.

THREE POSTS • AND WHAT A QUICK SEARCH FINDS			
#	THE POST	WHAT A QUICK SEARCH FINDS	RULING
1	Notice from the Corporation: garbage collection moves to 6 am from 12 October. Notice no. SW/21/2026.	The Corporation’s website carries notice SW/21/2026 with the same date and time. Two news outlets report it.	
2	Photo of a tiger walking down a school corridor. “TODAY in Hosur!! Share before they delete it!!”	The same photograph is in a news report dated 2019, about a tiger that wandered into a school in Assam. No outlet reports any tiger in Hosur.	
3	Researchers prove that drinking tea after 6 pm makes children forget what they studied. Forward to every parent you know.	No researchers, university or study is named. The same words appear on many parenting pages. No reliable outlet reports it.	

(a) Rule on each post in the right-hand column: *real*, *fake or invented*, or *real but misleading*. [3]

(b) Name the check that Post 2 fails, and the evidence that shows it. [1]

(c) Post 3 has been forwarded 4,000 times. Explain why that number is not evidence that it is true. [1]

6. Three students are making a video that will be shown free on the school website. For each, say whether they can use the material, and what they must do first.

(a) A song from a film released this year. [1]

(b) A poem by an author who died in 1920, whose copyright has ended. [1]

(c) A photograph licensed Creative Commons **BY-ND**. [1]

SECTION 3 – APPLY AND DIAGNOSE

7. On Monday morning a clinic in Mysuru finds every file on its main computer encrypted, with a message demanding ₹50,000 for a key. The clinic does have a backup, but it is on an external drive that stays plugged into the same computer all week.

(a) Name the malware. [1]

(b) The owner wants to pay “to be safe”. Explain why paying does not guarantee a happy ending. [1]

(c) Explain why the backup did not rescue the clinic. [1]

(d) Say where the backup should have been kept. [1]

- 8.** Another student has written two statements, and both are wrong. [4]
- (i)** “I deleted the embarrassing photo from my profile, so it is gone for good and nobody can ever see it again.”
- (ii)** “Trolling is when someone pretends to be someone else online, like the fake ‘Ananya, 13’ account that chatted to Meera for two months using stolen photos.”
- For each one, explain the mistake and give the correct version.

End of paper. Check that every defence you named comes with its reason, and every calculation with its working.